



RYLEE BARNETTE

SENIOR, UNIVERSITY OF CENTRAL ARKANSAS

CONTACT

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- 📍 Conway, AR

EDUCATION

2019-2023
Cabot High School
Distinguished Honor Graduate

2023-present
University of Central Arkansas
Biochemistry and Applied
Mathematics Major; Senior

SKILLS

- Spectroscopy (NMR, IR, UV-Vis) and titration
- Solution preparation and precise measurement
- Data analysis (Excel, R, Python)
- Quantitative reasoning and problem-solving
- Computational Software (Gaussian, Crystal17, Vesta)
- Organized, detail-oriented, and dependable

REFERENCES

Autumn Hurst

Arkansas Hospice volunteer coordinator
(479)-214-1120
AHurst@arkansashospice.org

Robert Mauldin

UCA chemistry professor
Research Mentor
rmauldin@uca.edu

Sanoj

Computational Chemist
University of Central Arkansas
Sanoj@uca.edu

University of Central Arkansas, Conway, AR

B.S. Biochemistry and Applied Mathematics | Senior | 4.0 GPA |

Relevant Coursework: Organic Chemistry, Biochemistry, Genetics, Analytical Chemistry, Inorganic Chemistry, Physical Chemistry, Physics, Calculus, Advanced Mathematics, Modeling/Optimization Mathematics

EXPERIENCES

ARKANSAS HOSPICE - VOLUNTEER

2023-present

- Provide emotional support and companionship to patients receiving end-of-life care, offering comfort, compassion, and one-on-one attention to those facing illness, fear, and loneliness.

PEDIATRIC COMPLEX CARE OF ARKANSAS (PCCA) INTERN

2024-2025

- Assisted therapists with pediatric care, supported children with severe disabilities in daily activities, and helped manage medical procedures such as feeding tubes and tracheostomy care while gaining hands-on experience in a clinical setting.

INFORMATION TECHNOLOGY (IT) STUDENT WORKER

2023-present

- Assist students and staff with troubleshooting, reimaging, and resolving technology issues, including help desk support, while developing leadership, problem-solving, and communication skills.

RESEARCH EXPERIENCE

Mathematics Research (2023-2024)

- Worked alongside the Math Department Chair to analyze Arkansas tax brackets and propose revisions to macroeconomic tax categories.

Cognitive Neuroscience Research (2024-2025)

- Studied ADHD, cognitive interference, selective memory, attention, and working memory, and presented findings at a research conference.

Environmental Chemistry Research (2025-Present)

- Conducting field and lab studies on longnose and alligator gar fish in Arkansas to measure mercury and heavy metal levels in water and assess environmental contamination.

Computational Chemistry Research (2025-Present)

- I conduct computational chemistry research focused on designing and optimizing crystalline structures for quantum computing applications. Using Python, Bash Script, and visualization software, I model structures that could generate entangled photons for use in advanced quantum computing techniques.